Model Monitoring

Model **monitoring** (drift, performance, and statistical views) is configured through the Model Registry API in `snowflakeR`. Monitoring requires **`snowflake-ml-python` >= 1.7.1** in the active Python environment (see `?sfr_add_monitor`).

You need:

1. A **logged model version** in the registry (`sfr_log_model()`, etc.).
2. An **inference log** table or view in Snowflake with a timestamp column and prediction (and optionally actual) score columns.
3. A **warehouse** for scheduled monitor computation.

Configure source and registry settings

```
library(snowflakeR)

conn <- sfr_connect()
reg <- sfr_model_registry(conn, database = "ML_DB", schema = "MODELS")

src <- sfr_monitor_source(
  "ML_DB.MY_SCH.INFERENCE_LOG",
  timestamp_column         = "EVENT_TIME",
  prediction_score_columns = "PREDICTION",
  actual_score_columns     = "LABEL",
  id_columns               = "USER_ID"
)

cfg <- sfr_monitor_config(
  model_name          = "MY_MODEL",
  version_name        = "v1",
  warehouse           = "ML_WH",
  aggregation_window  = "1 day"
)
```

Register and inspect monitors

```
sfr_add_monitor(reg, monitor_name = "MY_MONITOR", src, cfg)

sfr_show_model_monitors(reg)
sfr_get_monitor(reg, name = "MY_MONITOR")
```

Read monitor metrics

```
sfr_monitor_drift(reg, "MY_MONITOR")
sfr_monitor_performance(reg, "MY_MONITOR")
sfr_monitor_stats(reg, "MY_MONITOR")
```

Time ranges and granularity follow the function arguments documented in **?sfr_monitor_drift** (and related pages).

Segments

Optional **segment** filters narrow metrics to a slice of the population:

```
seg <- list(column = "REGION", value = "EMEA")
sfr_monitor_drift(reg, "MY_MONITOR", segment = seg)
```

Use **sfr_add_monitor_segment()** / **sfr_drop_monitor_segment()** to manage named segments on an existing monitor when supported by your SDK version.

Lifecycle and vetiver bridges

- Suspend / resume / describe: `sfr_suspend_monitor()`, `sfr_resume_monitor()`, `sfr_describe_monitor()`, `sfr_delete_monitor()`.
- **vetiver** interoperability: `sfr_monitor_to_vetiver()`, `sfr_vetiver_to_metrics()`.

Notebook example

The shipped Workspace notebook **workspace_model_monitoring.ipynb** walks through a full monitoring setup with `sfnb_setup.py`.

See also

`?sfr_monitor_source`, `?sfr_monitor_config`, `?sfr_add_monitor`, `?sfr_monitor_drift`, `?sfr_monitor_performance`, `?sfr_monitor_stats`